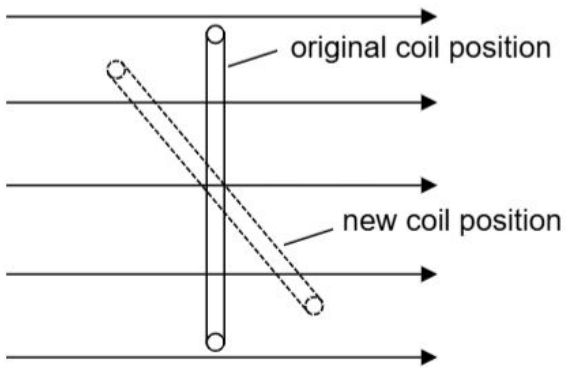
25 A coil is situated with its plane perpendicular to a uniform magnetic field and within the field.



The coil is rotated to the new position shown.

How does the magnetic flux density and the total magnetic flux passing through the coil change during the rotation?

	magnetic flux density	total magnetic flux
A	decrease	decrease
B	increase	increase
C	no change	decrease
D	no change	increase

Ans: C

Magnetic flux density: Unchanged (external field is unaffected) → no change

Total magnetic flux: Decreases from maximum → decrease